```
+----- Cursor Parking ------+
|     bean-tech_eranwt      |
|                           |
+---------------------------+
```

Computer Science Module Choice Opinions For 2024-25

THIS DOCUMENT IS OUT OF DATE - THE MOST RECENT VERSION IS HERE:
Module Choice Opinions For 2025-26

STOP TRYING TO EDIT THIS DOCUMENT PLEASE

What is this?

This document is a place for people to comment on modules running next year, so younger students can get advice from older students about what modules will suit you best. Think of it as a mini-Wikipedia for modules next year, but opinions as well as facts. Feel free to put whatever you want below, and also edit other pieces of advice to neaten things up. If you don't agree - don't just delete. Comment underneath saying you disagree or explain why.

Disclaimer: This is just crowdsourced so take the information with a pinch of salt; look at the department's actual comms and website about stuff for more information.

DO NOT ENTER INFORMATION ON PREVIOUS YEARS MODULES - ONLY ADD INFO IF YOU HAVE TAKEN THE MODULES THIS TERM!!!

Links!

- Module Choice Opinions 2023-24 (previous year's doc)
- Full List of Programmes (Select your year to see the options you have!)
- CSS Discord (Feel free to ask anyone questions about the modules here!)

Key

Code & Link	Module Titles	Semester 1 Credits	Semester 2 Credits
● Opinions			

Compulsory Modules

06-26581	BSc Computer Science Project	20	20

- Getting a good supervisor is 20% of it. If they're personally invested in your work, there's a much higher chance they'll actually care and put in extra effort to guide you.
- Worth looking at supervisors early, esp what they specialise in and how much their work interests you
- Start researching early, before the start of semester if you can, to give you a headstart for writing the proposal
- You need not take your supervisor's proposed projects if you have an idea that both of you are interested in.
- Study whatever necessary within summer break, it's life changing
    - Though this is not a requirement, to add to the opinion above. Don't stress yourself about studying during the summer, because projects can change a lot throughout the year. If you are studying for your project, make sure it's something you enjoy studying about otherwise you will just get burnt out
- Website tends to be easier, research can be fun and also killing at the same time.
    - The research part is definitely a good experience if you're doing it as a combination of other subjects (mine being human vision and cs)
- It's been useful for me to remember that CompSci Project is equivalent to two modules, it's not your whole grade. 3rd year is a balancing act and you're probably going to need to make sacrifices here and there, so keep in mind how much each thing is work before you sacrifice everything (friends included) to the CSP-gods
    - I second the opinion above, if you are like me who has taken a lot of exam-based modules, you might have a higher percentage of your final year grade come from the exams instead of the project
- It seems that what each inspector wants varies wildly. Some inspectors will give a lot of feedback, some only give a few sentences for your demonstration. Some say that you must focus on your evaluation/it's a key part, others will say that the evaluation doesn't matter and it's about the technical aspects.

06-26581	**BSc Computer Science Project**	**20**	**20**

- The report is the most crucial part of the project, a good report can save a project that is lacking
- Try not to compare yourself to other people; you are doing different things, more than likely, so it's apples to oranges. Just focus on the goals that you set for yourself and your deadlines.
- Try not to leave writing your report until the last week, it's 30% and the most crucial piece you'll hand in for your supervisor and inspector to understand your project completely.
- Here's some really cool advice for anyone doing one of these. Come up with an idea during the summer and get some opinions on it from some faculty members (basically a mini proposal). Then you can basically skip the entire planning part at the start of term and crack on with the project itself while you actually have free time to do it (I recommend living like a hermit for the first 6 weeks of term). Then, when they throw all the assignments at you, you can just put the project aside for the rest of the semester and pick it back up at Christmas

06-26587	**MSc Computer Science Project**	**20**	**40**

-

Optional Modules

06-35309	**Advanced Functional Programming** **Advanced Functional Programming (Extended)**	-	20

- AFP is quite different to FP, I'll give you that
- Instead of Haskell, it uses Agda, which is a similar language but is often used for proofs
- Vincent (PLPDI) has an Agda club in 1st Semester, if you're doing this I'd consider checking it out, I didn't get the chance to but it may help
- Like PLPDI, it is not an easy module, but it is incredibly rewarding when you get it right
- If you enjoyed FP and want to take on AFP, I'd certainly say go for it
- Agda is basically where all the theory and proofs from logic meets functional programming
- Would suggest to have a look on some Agda code in advance as the syntax is different from Haskell. If you could understand what the task is doing, then go for it.
- The lecturers are pretty nice. They deliver the material in an easy way to understand it.
- This is the module that will make you attend the labs and the office hours. Often, the provided solutions for the exercices are "too smart", which does not help with the understanding.
- A tough module but you can do extremely well if you understand the content (easier said than done). You have to be familiar with a lot of concepts from first year logic and TOC, and of course FP - it almost goes without saying but this isn't one to take if you didn't enjoy Haskell. Make sure to put enough time in to understand the content, complete the lab sheets/homework and definitely ask questions in labs or go to office hours if you get stuck. Also attempt some of the additional exercises in the handouts (be warned they are very hard).
- If you're doing Maths + Comp Sci, then this is a must. You're already doing proofs in other math modules and this is essentially the same but in programming

06-30209 06-38963	**Advanced Networking** **Advanced Networking (Extended)**	20	-

- Story time with Ian
- Loved it - it's not particularly "advanced" but still very good module if you're interested in network stuff at all
- Extended version is literally identical to the non-extended networking
- Please, please, please take this module, it is so good
- When your lecturer comes in after a concert in London, low on sleep, and delivers a banger of a lecture he wrote up the day before, you know you're in good hands.
- This is a good exercise in efficient note taking, and reading RFCs. There is a lot of content in the lectures, and if you try to write it all down you will struggle. As

06-30209 06-38963	**Advanced Networking** **Advanced Networking (Extended)**	**20**	**-**

basically all of networking as we know it is defined by RFCs, you will get a massive leg up on understanding if you read them and refer back to them in your notes.

- One of the easiest modules to take (comparatively) - it's mostly memorisation.
- You will learn key phrases such as "Don't do crimes", "What can possibly go wrong?", "Fix your DNS", ""
- Get ready for questions where you have to calculate distances and all of that stuff. They have a high chance of appearing in the exam, even if Ian says no

06-35308	**Algorithms and Complexity** **Algorithms and Complexity (Extended)**	**-**	**20**

- Mid term was hard asf
- The midterm was tricky, but you get a week to do it in your own time, and you can use the lecture content and weekly assignments to help you.
- I actually genuinely am really enjoying this module. The content is challenging, but doable and the material is interesting enough
- In terms of the module, it's really interesting! Rajesh covers about 6 weeks of algorithm paradigms: traditional ones like greedy algorithms, and divide & conquer, and modern paradigms like exact exponential algorithms for dealing with NP-hardness. Anupam covers complexity theory, like NP-completeness, and boolean circuits. There are also weekly assignments with a single 2% summative question to complete, again about a week to complete it, which really help with understanding. The weekly assignments are also accompanied by another 3 or 4 questions as extra practice that you don't need to submit, but are great tools to help ensure you understand concepts well.
- You get a much better understanding of the complexity class NP, what it means to be NP-hard, and NP-complete. You also get familiar with the idea of creating polynomial-time reductions to prove NP-hardness, which is super cool.
- There are some cool mathematical concepts like proving the closure of the class P under operations, and you do a lot of algorithms on graphs so you become very familiar with graph notation and concepts.
- This was the module I had the least confidence for but since starting, it's now top 2 for ones I'm most confident for. Content basically builds on everything you've learned about (classical) computation theory and has you applying it to design algorithms, analyse them, etc. The weekly assignments force you to stay on track to at least some extent. All assignments up to week 7 I've found very doable, including the midterm given we had a lot of time to go through it and come up with thorough answers (however still no marks back at time of writing).
- The difficulty for each of the assignments has been mostly non-linear. It's a complete gamble on whether or not it will be easy or hard this week. I remember Week 7 being easier than week 4
- Minimum Vertex Cover mentioned. 10/10

06-35308	**Algorithms and Complexity** **Algorithms and Complexity (Extended)**	-	20

- I have no complaints about this module. Good pacing, good structure, good content, reasonable difficulty imo
- A solid module - it's important to keep up with the content though.
- Make sure you are good with your graphs

06-30211	**Computer-Aided Verification** **Computer-Aided Verification (Extended)**	20	-

- I don't know who deleted my feedback on this module but I hope your pillow is warm on both sides when you turn it over
- The feedback for this last year was quite poor, but the module did run a lot better this year
- They kind of came out of nowhere with all of the formative assignments, but otherwise it's good to practise with for exam preparation so not a big idea
- The labs were rather useless. They were in the UG04 at the same time where other students were present doing their own work so it was really hard to understand the TAs
- Sometimes when ppl asked the TA questions, he would reply with "you should know this" or "i can't tell you the answer" (like he knows the answer but doesn't want to tell us)
- Instead of SPIN and Promela, nuXmv was used, which is ancient as dust but has pretty good documentation + it was taught pretty well in labs
- Pretty much Theories of Computation on steroids

06-32167	**Neural Computation** **Neural Computation (Extended)**	**20**	**-**

- Overall, Neural is a pretty well put together module; the content is all there with recorded lectures, slides, weekly Python labs, and extra reading too. Some of the later weeks' content is challenging and definitely takes some work to fully understand but it's very interesting and worth the effort. The 20% CA apparently used to be a group project to create a neural network, however, this year they used 2 10-question Canvas quizzes instead. Unfortunately, I find some of the content is taught a little dryly in the on-campus lectures.
- I second the opinion above, the lecture slides are the only things that are useful, really.
- The new 10-question assessments are a bit annoying. They are mostly multiple choice trick questions, so have fun revising the entire module so as to not get caught off-guard. The easier questions on Quiz 1 ended up not being marked because they were unanswerable.
    - In spite of this, this module is very rewarding, because the field has grown rapidly in the past couple years, so it is at the forefront of AI. It powers all the new AI art that you see, as well as ChatGPT
- The lab exercises are also pretty cool. They give skeleton code that you complete to create different forms of neural network. You get to generate handwritten digits!
- A nice companion piece to Machine Learning, because some of the content in both of the modules is shared (Gradient Descent). Machine Learning is very much the "how" of AI and Neural Computation is the "what".
- Well structured module. Interesting content. Module team is very helpful too. It's worth mentioning that the lab Python notebooks with the skeleton code have text summarising the week's content so you don't need to go back and forth between slides and code. Makes the notebooks really helpful in reinforcing already learned content. Genuinely looking forward to revising the content before the exam!
- Before you pick this module, choose a side: are you y-bar or y-hat?
- I enjoyed the content and the labs - you get to train and play around with most of the neural networks covered. I would suggest reading the recommended reading as it explains most concepts better than the slides and the lectures. The lecturers are not brilliant but again the content is good, and the module well-structured, so it's not hard to stay motivated.
- Unfortunately, I deeply regret choosing this module. The teaching and slides were useless and I found myself relying far more on YouTube videos and books. The exam and the quizzes were so obtuse and badly worded that I feel that it's going to be the only module where I have a realistic chance of failing it. I feel like the module was a waste to choose as I could've learnt everything myself without tanking my grade.
- The exam they gave us was incredibly lazy - it had an identical question from the 2022 exam, creating an unfair advantage for those who had got hold of the exam (they didn't give it to all of us, it was leaked).
- Super interesting module, would take it again. However, the exam was badly written (not all of it was even written lol). It focused on only the last 3 weeks if I recall correctly.

06-30213	**Computer Vision and Imaging**	**20**	**-**
	Computer Vision and Imaging (Extended)		

- The following feedback details a tragedy in three parts:
- Very good module with lots of interesting content
- Hamid is a great lecturer and he even manages to finish a lot of lectures early
    - Those lectures were a bit too fast-paced imo
- There are weekly unassessed programming tasks for the first few weeks where you get to implement a bunch of image processing algorithms (edge detection ,gaussian blurs, etc)
- After this, there is a **3 page project write up worth 50%** which uses a lot of the algorithms you implemented in the optional programming tasks beforehand
    - I advise you do the optional tasks as it will make the final assignment much easier
        - +2
- I think it is a lot easier if you have done neural computation the previous semester as there is a lot of overlap and content like convolution is explained a lot better in neural comp.
- The feedback on the weekly unassessed programming tasks can be a bit shit depending on which TA marks it, especially as that feedback would be helpful for the final assignment. Otherwise, a really interesting module! Hamid speedrunning lectures in like half an hour is great.
- Hamid can be harsh since he speedrun all the content, he doesn't allow you to think during and when you realise the lecture is already finished. However, the weekly lab does help a lot for understanding the content.
- GET RID OF THIS DAMN 3 PAGE LIMIT, IT'S NOT COOL
- The beeee

06-30213	**Computer Vision and Imaging** **Computer Vision and Imaging (Extended)**	**20**	**-**

- I think it was missing some two-markers
- There were two-markers?

- Very interesting module but very strict marking lol.

06-38963 06-38966	**Dependable and Distributed Systems** **Dependable and Distributed Systems (Extended)**	**-**	**20**

- Storytime with Matt! Genuinely just an excellently taught module. Matt responds to emails basically instantly at all times of day (including 2am), and enjoys answering questions. Full powerpoints & videos on canvas. Coursework is a 2000 word literature review of a dependable/distributed system of your choice, worth 20% of the module. This year Matt offered to read and review drafts of every student's work before submission, which was great for final improvements!
  - Re: the coursework, don't let decision paralysis make you wait 2 weeks until the deadline before you start. There is massive flexibility on the actual topic of the review, so don't be afraid to pick something you really enjoy thinking about.
- Probably the best-taught module on the whole degree. Content won't be everyone's cup of tea though.
- This was an incredibly well put together module! I picked it as my last-minute choice and it ended up being my favourite so far. Matt is great, his content is great, his feedback is great; everything is just really great. There's a good range of content from dependability, to distributed algorithms, to clocks and broadcast algorithms. It has some other aspects sprinkled in like complexity, maths, proofs, etc., though he tries to make the module less about formal verification so it's very accessible.
- If you are not a massive fan of maths, you will likely be in good company as Matt doesn't like it either. You will come across some scary looking maths, but it will be very accessible.
- Attend the lectures, and arrive on time, or face disappointment.
  If you've got a good reason not to, Matt will give you all the exam advice he gave in the lecture over email.
- Very well taught, as much as I disliked doing the coursework it helped a lot and makes doing a lit review for your project so much less daunting.
- Personally I found the literature review quite interesting as I had full freedom to pick my own topic and the points I was going to convey, Matt's proof-reads of my draft helped a lot too
- Matt is incredibly passionate, approachable, responsive and helpful. Personally I'm not the biggest fan of his lecturing style (he repeats content multiple times rather quickly rather than going through it once, but slower, and also taught new content in weeks 6 and 11) but I guess it just comes down to personal preference. Content is a bit dry but he makes it entertaining.
- I think this module runs in second semester next year

06-38963 06-38966	**Dependable and Distributed Systems** **Dependable and Distributed Systems (Extended)**	-	20

- It does - I've amended the header
- There is zero machine learning in this module. If you don't respect AI, do this.

06-35310	**Evolutionary Computation** **Evolutionary Computation (Extended)**	-	20

- Shan
  - He
- Comedic Gold
- Somewhat tame module with interesting assignments
- Not the best but because I also take mobile the bar is already on the floor
- You can choose any language you want for the assignments - the recommended for Assignment 1 is Matlab (can just use Python tbh), and Assignment 2 recommends Python/Java with the additional use of Docker
  - You like writing long reports on what you have done? Good
- Not sure how practical what I have learnt is, but it's all quite interesting and assignments are good
- The second assignment is awful, almost all the problems I had were due to docker, it makes it extremely difficult to bug fix anything.
  - It is essentially a test to see if you can use docker.
- I literally cannot express how much I hate the second assignment
- Really good module in terms of the lecturers (especially shan he) and the marking.

06-30214	**Human-Computer Interaction** **Human-Computer Interaction (Extended)**	20	-

- Often considered "the easy module", which is kinda true. I'd say it's **easy to pass, but difficult to get a high grade in**. It's 100% a team project - so entirely up to the strength of your team. Coursework is producing a 100 slide presentation on how you solved that year's design challenge + a short solo essay talking about part of your process. Our prompt was "Create something that helps people contribute towards addressing climate change", and the year before was "Design a digital technology solution to improve people's experience of cycling", so I expect next year's prompt will be somewhere between those two in terms of scope and vagueness. Russell is a fun lecturer, and he gives good advice. A great module if graphic design is your passion!
- You get to interact with others in the early years to then make your own teams.

06-30214	**Human-Computer Interaction** **Human-Computer Interaction (Extended)**	**20**	**-**

When you're on good terms with yours then you'll find it pretty fun!

- Make sure you put the effort in.
- Not very *difficult* but takes a lot of time. Make sure you're actually working on it every week
- k (1-2 meetings every week is probably fine) otherwise you're not getting a good grade.
- I would recommend making sure you known at least one other person doing the module who you work well with, to give yourself a leg up when assembling a team
- Make sure you are confident that you will work well with your team, as you have 4-5 weeks to try out different groups to see who you work best with. A conflicting or unresponsive team can cause unneeded stress
- Russell allowed us to book 15 minute online slots where we meet on zoom for feedback on our slide decks, as well as sitting with our group in person to provide very detailed constructive feedback, which helped our slide deck a lot

06-38964	**Intelligent Data Analysis** **Intelligent Data Analysis (Extended)**	**20**	**-**

- Lecturer Peter Tino ensures that you understand the content (Principal Component Analysis and document mining).
    - Goes deep with first principles to lay a foundation of why we are doing what we are doing and how it all works
    - This also makes the assignment decently easy if you've been listening
- Second half is Felipe Orihuela-Espina.
    - Engaging.
    - Will make a point about engaging with the content and that the stuff that is taught is to get you thinking a certain way and give you the
    - Building blocks to come up with solutions to new problems.
        - I.e. think about application of content
    - Have you heard of the 7 bridges problem? You soon will :)
- Very good module 9.5/10
- Examples given in Matlab though I believe you can use any language
    - Tutorials get you to translate the Matlab into another language
- Good module. The buildup towards PCA was quite good, with all basics covered. Helped me grab an intuition before implementation.
- The labs included QnA, small quizzes to test knowledge from the lecture, and demos of real world usage/examples, which were really interesting.
- Peter Tino is so nice and considerate. Felipe's lectures were some of the most mentally stimulating lectures I've had at Uni, the guy is actually inspiring. Special shoutout to Marco Canducci for running the labs and

06-38964	**Intelligent Data Analysis** **Intelligent Data Analysis (Extended)**	**20**	**-**
demos!			

06-37809	**Intelligent Interactive Systems** **Intelligent Interactive Systems (Extended)**	**-**	**20**

- A simple, interesting module. Two assignments: a literature review and an essay, worth 20% of module grade (10% each). In our year half the module was taught by Mark Lee and the other half was taught by Mubashir Ali. I found Mark's content way more engaging - lots of talk about ethics, game theory and ebay scams. Mubsashir's stuff is mostly reteaching AI1 content.
- Stuff about game theory, human decision making and fairness/bias was interesting. First assignment about that and interesting to research as well. Second half goes over lots of different AI models, how they work and how we use them - it's ok, but not as fun to learn about
- **There is required reading** and both the assignments are essays. We've been told the exam also contains long answer questions. Hope you're comfortable with the English language. As a dyslexic I've found it's a lot of work and hard to stay on top of all the reading, but doable (and helps when it's fun to read about interesting things).
    - This module is the first time I read serious academic papers for fun and not because I will actively lose marks if I don't.
- Dig up your notes on Statistics. It ain't heavy, but you will need it towards the end of the module.
- Mark Lee is an engaging lecturer
- If something in the lecture comes up that is surprisingly specific, note it down! Even if it didn't get more than a couple sentences on it. It may (did) come up in the exam.
- Get good at handwriting. They like putting chungus 10 markers in the exam that have you explore ideas in detail.
- Don't be surprised if something comes up in the exam that you don't get taught
- God bless Mark Lee x3

06-30227	**Intelligent Robotics** **Intelligent Robotics (Extended)**	**20**	**-**

- Be mindful about taking the Extended and Non-Extended versions of this. Non-Extended is 100% coursework, whereas the Extended is split 50/50
    - To add to the above, you do the same coursework as the non-extended version. So you do the same coursework, but it's worth half as much, on top of the exam in the summer.

06-30227	**Intelligent Robotics**	**20**	**-**
	Intelligent Robotics (Extended)		

    - If you're doing MSci, and want to do this module in either of your 3rd or 4th years, I'd recommend doing it in your 3rd year, as this is potentially one of the nastiest changes from a base module to the extended version.
- This year, there were three assignments. Two group assignments and one paper review.
    - The final assignment being a painstaking project where you implement whatever you learnt throughout the semester.
    - You'll demonstrate assignment one and three, everyone in your group should be able to answer any questions they asked.
- To be fair, the lectures were interesting but I felt like there is room for improvement in terms of how they are delivered. We finish one content and then we jump to the next one.
    - They're more theory based rather than implementing them.
    - Why are there five hours of lectures????
    - If you do the exam, derivation of some equations are a must know which aren't covered properly in the lecture. The textbook has them.
        - The slides are also based on the textbook as summarised versions.
- **VERY** time-consuming, be prepared.
    - Late night work here they come :)
    - Be ready to put 30+ hour weeks in for the last 4 weeks before submission
- Content heavy, with the side of mathematics involved.
    - They aren't exactly easy to wrap you head around
- **Not an easy module to take.**
    - **+5**
- Mohan is quite knowledgeable in this subject but he can also be strict on some things.
    - ***Mohan is leaving by the way***
    - He's the only lecturer, aside from the Guest Lecturer who we saw once.
- We work with simulated robotics more so than actual physical robots.
    - This is dependent on if you want to and can work with physical robots for your final group project.
    - You'll need to learn how to use the Robot Operating System (ROS) quite well by yourself to understand how things like Topics, Publishers, Services, and Subscribers works.
        - In Python
- Keep in mind that unless it is changed next year you will "learn" ROS 1, and the robots you can use for assignment 3 run on ROS 2. This means that if you want to use a robot in the robotics module you will need to learn it all over again as ROS 1 and 2 are quite different.
    - To add this, This year the robot was computable with Turtlebot 3 and ROS 2, however on some Linux versions it's not possible to use either. If you don't want to suffer from dealing with versions and VMs, **Don't use the robot.**
- The lectures are almost if not entirely theory based and it will be up to you to learn how to actually implement said theory. A pdf is provided teaching you how to do the basics in ROS, however it is VERY limited and only covers the most basic of tasks.
- A VM or dual boot is needed for this module as ROS runs on Ubuntu (specifically and only 20.04 for ROS 1). I would recommend a dual boot as ROS is a very intensive program. **A dual boot/vm session is available at the start of the academic year run by AFNOM if you are unfamiliar with dual booting.**

06-30227	**Intelligent Robotics** **Intelligent Robotics (Extended)**	**20**	**-**

- **Dual boot is unavoidable.**
- The course is effectively an any% speedrun of robotics with 5 hours of lectures per week for the f rst half-ish of the semester before reducing the hours to 2 per week as all the content has been taught by then.
- You can pick your own team of 5 so make sure to pick people that you know will do the work as the assignments (especially assignment 3) will require everyone to work very hard to get them done.
- Terrible module, the content is ok, but the project is just stupid, asking us to make innovation in robotics, it's just like a final year project, except it takes up most of the module marks, while you also have the real final year project to work on. Our group worked very hard on this, putting a lot of time into it, but we ended up getting a very harsh grade, all that time wasted for nothing. The guidance is terrible too, the TAs don't really know what you should do in the project, and don't give good advice. I thought 100% continuous assessment meant no need to worry about the exam, but I wished I had taken another exam.
- Interesting content and the main assignment (team project) allowed you to choose pretty much anything related to robotics which was neat. **But** be prepared to spend every waking hour working on this module for the entire semester. I spent more time working on this than FYP in first semester.
- To quote last years feedback:

06-38965	**Machine Learning** **Machine Learning (Extended)**	**20**	**-**

- Fairly difficult module, heavily maths-based although you don't need to fully understand all the derivations (partial derivatives etc.)
- Two lecturers:
  - Leandro is really good in my opinion, goes at a good pace and explains things clearly. He's also responsive on
  - Teams and gives meaningful responses, and good communication in Canvas announcements etc. Clearly cares about students
  - Jian is ok, but goes pretty quickly and doesn't explain things as well/clearly as Leandro

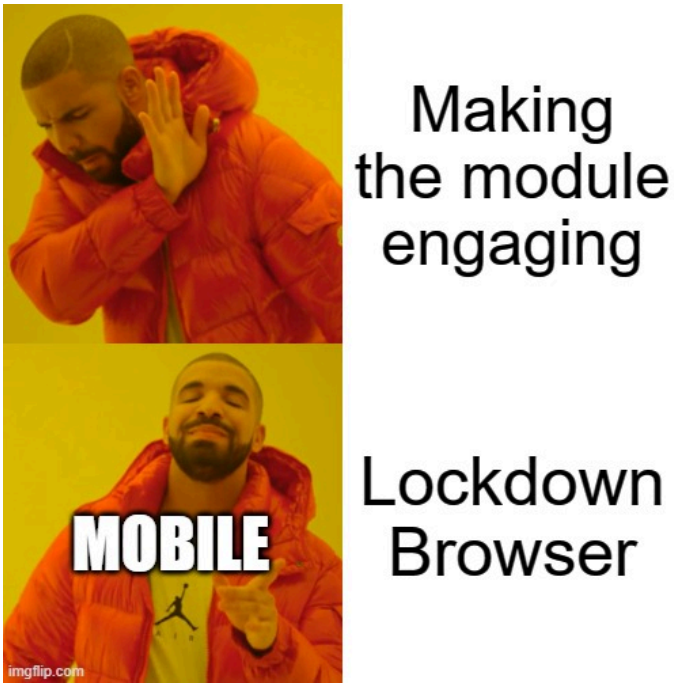
06-38965	**Machine Learning** **Machine Learning (Extended)**	**20**	**-**

- First half is mostly focused on actual machine learning algorithms/methods, how they work, and the maths behind them
- Second half is a lot more theoretical and less tangible content
- It feels to me like they stuffed too much content into the second half of the module - some of the later lectures took me a while to go through and properly understand everything
- Coursework is two quizzes worth 10% each, with a 1-hour time limit but you can take them at any point in the week. They test understanding and ability to apply formulae - they're not too bad if you complete the two unassessed exercises every week. I think the quizzes for the extended version are different in one or two questions
- Only content from the first half was actually assessed in the quizzes, so it's hard to tell how much of the second half content is important or what kind of questions there might be
- This year it was split off from Intelligent Data Analysis, and it kinda shows that they replaced that module's content with what feels like filler.
- Requires you to be very good in maths, especially linear algebra.
- Math lecture, will have to see what the exam is like
- Super interesting module if you're interested in AI/ML. Exam wasn't the worst, it was pretty nice :)

06-29289	**Mobile & Computing** **Mobile & Ubiquitous Computing (Extended)**	**-**	**20**

- Fuck Mobile. It is boring. It is stupid. It is useless. The quiz was broken. ARRRRRGGGGGGHHHHHHHHHHH
- **^based**
- Genuinely, the module is really poor. This year, we are assessed by two 15% open-book Canvas quizzes, a 20% write up, and a 50% write up. The write ups are about a concept for an IoT device, including things from SEPP - i.e., boring diagrams and useless thinking. Only Wendy seems to care about the module; Chris doesn't do anything outside the lectures and we've not met Abhirup yet. The lectures themselves are mind-numbingly dull - they take semi-interesting concepts and make them extremely boring. Some of the content taught involves Arduinos, Flutter for mobile apps, and TinyML for machine learning on mobile devices. However, NO programming is assessed. They say you can add it into your write ups for more marks, however, it's not a requirement and you won't lose marks for doing so.
    - The first quiz had numerous issues that made it stressful at first - images not loading, RAP not added for students who needed it, almost requiring you to install the shitty lockdown browser that is not available on Linux, surely someone has more

06-29289	Mobile & Computing Mobile & Ubiquitous Computing (Extended)	-	20

    - To fix it, they had to take it down, but once it was back online, they kept incorrectly changing the due date, preventing people from doing it at some times
- The module is known for being easy - it's definitely easy to pass, but not at all easy to deal with for an entire semester. You're much better off picking a module where the content won't destroy your brain cells, and where you can actually enjoy dealing with it. I've had friends who picked the module because it was easy but they're deeply regretting choosing it.
- Is it really that bad? I mean I just want some easy grade, and this module feels like an easy grade, yes the content is boring, but then I don't need to study this, and can focus my time on my final year project. The assignments don't require too much of your time too, so I think it's just an easy module.
    - That would be up to you. Choosing a module is down to only you, and if you are happy to not go to lectures, not read the content, and just look things up when assignments are released, then you would probably do that. However, I gave feedback from the point of view that you want to make the most out of your degree in terms of having interesting modules and actually learning.
    - "Just look things up" may be putting too much trust into the module team following what the Internet may say-
- Lmao yeah fuck mobile, it's ass - if you don't miss the piss poor management of OSSP, Team project or SEPP, for the love of god, don't pick this
    - +1
- 

- Generally easy, but so far it's been difficult to find motivation to get into the actual content. Makes lectures, quizzes and assignments a drag. Also yeah I'm surprised at how poorly the module is managed, given its popularity. The slides are very

06-29289	**Mobile & Computing** **Mobile & Ubiquitous Computing (Extended)**	-	**20**

bloated too, they should probably get rid of some of the absolute bs diagrams.

- Wendy is very responsive to questions and provides good constructive feedback, seems like she really cares
- Dogshit
- If you want a module lead (Christopher Baber) who responds sarcastically as fuck, pick this module. “I note that as you were writing your email, it perhaps became apparent why you did not score full marks on each of the marking criteria. We believe that the marking scheme has been applied appropriately.” - they did not read my report or apply the marking scheme “appropriately”. MF clearly doesn't have etiquette so don't expect anything reasonable from him or this shit module .
    - A concept that emphasises the use of blockchain, the use of a microcontroller to automatically analyse sensor data, and the use of an RFID sensor to identify vehicles uniquely which can be used to view a vehicle's maintenance logs stored on a blockchain was deemed as “not novel”. I clearly needed to take fucking shrooms or smoke some crack and go on a trip to another dimension to come up with something more novel - my fault I didnt.
    - Probably the least professional lecturer in the school
- Easy but badly run module. Expect unprofessional feedback from lecturers bordering upon rude. Also don't expect your assignment to be read properly when they mark it, a good majority of the year got marked down for things they needed to include when it was quite clearly there. I.e. not having 3 diagrams when 5 have been included. Don't expect any useful feedback, and don't expect to get your mark changed, they don't even want to have a conducive conversation about why you got the mark either.
- If you don’t care about anything except your degree classification, then pick this module. Even though it is badly run, you’re only going to do two assignments and two quizzes; no exams. So overall, one of the easiest modules compared to other options.

06-37810	**Natural Language Processing** **Natural Language Processing (Extended)**	-	**20**

- Note: NLP requires you to have taken Neural Computation.
- I feel like it has some interesting content, but it can feel a little dry at times. Phil starts off with teaching basics of NLP and text processing, as well as basic AI models, and then Venelin takes over and teaches Neural Computation Part 2, just relating it to NLP specifically. There was 20% in-person closed-book assessment in the labs, that was alright.
- I'd say the lectures are alright, but I don't like them only showing the PDF with the model answers on projectors in their lab sessions, and not uploading them on

06-37810	**Natural Language Processing** **Natural Language Processing (Extended)**	-	**20**

Canvas. So basically if you want to revise the content later, you have to take a picture of them and this is sometimes horrible especially when the question is about implementing some pieces of code.

- Phil's half is pretty straightforward but has a lot of content - I recommend making an effort to stay on top of things. It is a bit frustrating that so much time is spent on traditional techniques that are no longer used. Also the quiz was a bit annoying as some of the correct answers seemed very subjective, and they only told us the format a week beforehand. There are weekly practice/lab exercises, but solutions are only provided in the labs and half of them are wrong or missing, which is annoying. Also there is no assessed programming, despite there being 4 hours of labs a week. Venelin's half is a lot more interesting so far - he's an incredibly knowledgeable and passionate lecturer so I'm enjoying the module a lot more now. Overall I would recommend taking it if you are interested in NLP as it provides a really solid foundation if you want to work in the field in future.

06-30230	**Programming Language Principles, Design, and Implementation** **Programming Language Principles, Design, and Implementation (Extended)**	**20**	-

- Challenging, but taught and assessed very well. The module team are genuinely passionate about what they teach (shoutout to Vincent in particular!)
- Each lecturer uses slightly different notation; switching between them can be annoying, especially when you're just starting to grasp the content.
- Vincent was epic and the content was fun
- Vincent makes everything better, in particular a module that is already super interesting. If you liked the reverse engineering from the security modules then you'll love this. Also, the unexpected return of propositional logic for the first assignment so if you love solving those big old tree things from the first year, take this module.
- I'd definitely say take the module if you're interested in the content behind it, it's very difficult but very rewarding because of that
- Take PLPDI, pros: Vincent and Sean, cons: Nothing
- Next year, it says Professor Uday Reddy is teaching it.
    - He taught it this year too! He took over a couple of years ago. He's just not on Vincent's or Sean's level
- Vincent is really good at explaining stuff.

06-30231	**Security of Real-World Systems**	**20**	**-**
	Security of Real-World Systems (Extended)		

- Interesting and engaging content with two great lecturers
- Well-paced in my opinion
- If you don't like low-level things like assembly, machine code, raw data, hex etc this module isn't for you
- Coursework is only worth 20% which isn't great considering the amount of work the hacking assignment takes and how practical the subject is
- There's a group assignment to hack a game, which is fun in theory but in my experience it was far more effort than it was worth (only 8% of the module). Pro tip: don't pick a game that nobody's hacked before because it's pure pain. Also pick a group where you know everyone's competent and will put in the effort. And start working on it at the start, not halfway through the assignment.
    - To add to the point above, next year they may have a game designed that can be used for hacking (Project Pwnie). They didn't quite make it into a ready to use state this year, though.
    - Our team tried hacking that game anyway… it was a struggle
- On the flip side, the human security exercise was great fun and goes against everything you've ever been told at school and university. Probably the funniest thing you'll get to do as part of your degree (if they do this exercise again)
- Definitely the most interesting module I've done so far, like others have said the game hacking assignment is a lot of work for the amount it's worth, it is quite fun though but can be a bit stressful as you don't have long to do it.
- A rare module where there is actual programming involved and it's a big part of the module, and not just as unassessed exercises
    - I actually disagree, modules like OOP and FSD have lots of assessed programming and this has no where near as much as those. This is unique though as programming in this module is really used as a tool to practise/apply the content without the content itself being programming which you might enjoy if you don't like just churning out code and want to see something practical in this field. The game hacking assignment does have quite a bit of code but you're probably spending more time reverse-engineering and figuring out how to do things than just writing lots of simple code or algorithmically-complex code.
        - To be specific, I was talking about 3rd year modules
- Tom and David are the lecturers for this module. Tom covers the binary analysis and exploitation, whereas David covers the side channels and fault injections (breaking into hardware and encryption).
    - Just as previous years have said, they are both excellent lecturers. Not only are they passionate and very knowledgeable about their field (they provide the scientific papers and further reading that back up their teaching), they are just great teachers who can communicate these topics well.
- I've disliked every module I've done for the past 2 years. This module was the first module I genuinely enjoyed and would definitely recommend it if you think you'll be

06-30231	**Security of Real-World Systems** **Security of Real-World Systems (Extended)**	**20**	**-**

interested in it (and I'm most interested in writing software). It's not easy or free credits and the group project will take 20+ hours despite being worth 8% but in the end on the whole it's a solid module. Tom's amazing and David is knowledgeable and has actually worked on the stuff he's teaching which helps you take it more seriously. Even if revising for the exam will be stressful when we get past questions (they don't provide much exam context in lectures and I actually have no idea what they would ask about David's content since it's basically just a bunch of examples of attacks which are interesting but not really typical exam question-style content without being a big memory test), this module was worth it. Do not delay the group project, work on it as soon as it's set or you're going to have a bad time if you want a good grade. Also seriously think about LD_PRELOAD vs DLL injection, one is *a lot easier* than the other. David's content gets a bit boring after a few lectures as it's just attack example after attack example after attack example but he's a good lecturer and some of the ideas that went into the attacks can be interesting

- Very interesting but tough module, will require you to do a lot of self learning to get good at the content and get a high mark in the exams, don't see exams being any easier imo
- The game hacking assignment is probably the most fun assignment I've done on the course, but at the same time the most time consuming for how much it was worth. It was only worth 8% of the module, although I easily spent dozens of hours on it. I would highly recommend it if you enjoyed security and networks - please be warned if you didn't as there is a LOT of assembly and low level attacks. You basically have to be able to read and understand assembly code.
    - I'd say take this if you enjoyed security but **beware** it's nothing like the second year security module. You can think of it as a continuation of the second year security module where you start with buffer overflows and low level stuff (like in the last part of S&N).

06-34240	Secure Software and Hardware Systems	-	20

- Great module overall. It is now 20:80 continuous assessment:exam compared to previous years where it was 50:50. This year, teaching was split between Mihai Ordean, Marius Muench, and David Oswald. IMO this was good - we didn't spend too long on one particular area and it helped keep me engaged throughout the semester. The assignments have been really interesting, effort has clearly been put in to make these more engaging and give some experience in practical security. Lecturers clearly care about the content and helping students understand it. Also, this is a Masters level only module, so can be taken in 4th year of MSci or on the postgrad masters. I would recommend doing Security of Real-World systems in 3rd year if you are an MSci, as some content is referenced or repeated.
- Mihai went very quickly through his lectures in his earlier weeks until some of us emailed him complaining that we did so, after which his delivery of the lectures was of a more reasonable pace. If he does this again next year just shoot him an email and he will slow from Red Bull to Alpine

06-30208	Teaching Computer Science in Schools Teaching Computer Science in Schools (Extended)	20

- This module runs over two semesters but counts as a second semester module as you will do the training and DBS Checks in the first semester and then your placement in the second.
- Location is not limited to Birmingham. You can work in places you've gone to or know about as long as you let the module lead know about your interest.
    - You'll probably need to do a week and a half instead of eight weeks depending on how much you're willing to spend.
        - Cost is covered DEPENDING on the case. If it's like a short distance, then that's fine.
- More focused on Primary and Secondary Education but you can do your placement in Sixth-Form too.
- Bob took over the module this year for the first time and it's clear he's put in effort to make the module good for us. The module had a lot more students than previously and he managed to find extra places to accommodate more of us, which was really good of him. There are guest speakers that come in to talk about their experiences and it's great to see just how passionate they are about their jobs and teaching.
    - Bob is great! He'll support you whenever he can 😄
- You will do a total of **eight days** placement if you choose to do the expected route. You'll be shadowing, assisting, and finally teaching your own lesson.
    - There is an option to either run an after school session OR teach!
- Should be an awarding module! A definite change of pace from the standard CS teaching. You'll get to inspire and motivate younger students who might have the knack for Technologies.

06-30208	**Teaching Computer Science in Schools**	**20**
	Teaching Computer Science in Schools (Extended)	

- Now that the placements have started for us, albeit late, it is sooo much fun to go to a school once a week and experience what it's like. To be able to meet different teachers and understand their teaching styles, their approaches to engagement and behaviour, their lesson planning, etc., it's all so interesting. And being able to see how you participate more and more each week is super rewarding; we started off simply sitting in the classrooms and watching in, but now we've helped individual students, walked around the class to observe, made starters, and ran simple activities, as well as leading the after school robotics club! Each placement is different too, so it's great to chat to others on the module to hear about their experience and what they've enjoyed.
- Assessment involved a weekly 1-page report detailing your activities at the school during the week and an evaluation, then an overall feedback report from the school themselves assessing your performance and attitude, and finally a poster in your pairs to talk about your experience in the placement.
- First semester: weekly lectures from external speakers covers the "content" (educational/teaching theory, why teachings are worth doing, etc). There was an assignment getting us to research some area of teaching/educational technology and present it which was cool. This semester is pretty chill but make sure to engage with everything so that you do get allocated a placement (they may kick un-engaged students of the module if there are not enough placements)
  Second semester: **8 days of school placement** typically done one day a week. This has been an amazing opportunity and would definitely recommend it even if you're not interested in teaching. It's so different from anything else you'll do through modules and you can learn a lot. On the flip side, waking up at 6 and commuting is not fun and there aren't always things for you to do (some days were spent observing causework lessons where we were not meant to get involved). Even then most of what we've done has been interesting. Experience may vary depending on the school.
    - I would agree here, being placed in a class where it isn't really CS related and a coursework doesn't offer a lot to learn. Aside from that, it's quite enjoyable! Might be disorganised at first but once you get into it, it's hard not to fully hate the module.
- **Think about this as a semester 2 module**. That's where the bulk of the work is. If I'd know that I should have picked other modules differently to make sure I wasn't doing a 1:3 split.

Further Advice

For Everyone

- I'd advise people to consider doing a 3:1 split if you can, because it's not likely that you get a lot of your project done in the first semester. If you're a procrastinator like me, then it makes a lot of sense to at least get an extra module out of the way
    - Probably also makes sense to do this split if you have more exam-based modules. Since the exams are all now in the summer (after doing the project), you really only have to worry about revising when you have finished your project. Not ideal, but it is one way to min-max your grade
    - I did a 2:2 split and was still 2 weeks behind at the end of semester 1 and barely touched my project, so I definitely wouldn't have had time for a 3rd module. I guess it depends on which modules you do and how you manage your time
    - I did HCI and Security of Real-World Systems and the 2 projects at the same time took a lot of time and made it hard to work on other things at the same time, but I do think 3:1 is worth it as semester 2 is already going to be chaotic and you'll be able to catch up on your project especially with only 1 other module
- Don't think that doing all coursework modules will be easy. Trust me, it will take a lot of time to get through everything as you'll need to work with groups very often to complete one task.
    - If you're up for it, then feel free! Just remember to make time for the other modules as it can get time-consuming and daunting if you leave it last minute.
    - Second these 2 points^^
- I do agree with people suggesting taking 3:1 but don't think it is possible when choosing Intelligent Robotics (take way more time than you may think it will). Also, don't do 3:1 if you don't like the subjects you have to choose, better do 2:2 and study what you are more interested in.
- Keep in mind that dissertation submit day will be like a month BEFORE exams. So think it through when choosing both subjects in the second semester which are 100% continuous assessment.

For Maths and Computer Science Students

- Consider doing AFP as this module involves doing proofs and basically just logic which is essentially what all maths modules do

For Degree Apprentice Students

- DA requirements can be found here.
- Vincent is a king. Wendy is a queen. They genuinely care about you doing well, and will make time for you if you can't make it to one of the advice sessions, but still want some advice.
    - Do show up to them. Even if it's just to hear 'yeah you're doing good, do more of that'.
- Don't stress about the Gateway or EPA. The Gateway is basically a box ticking

exercise. The EPA is FYP Inspection: The Sequel. If you did well in that, just do it again but with more focus on the Competencies and Specialisms. Don't forget to talk about how your project helps the business.

- If you are on the PwC course (has Vodafone fully pulled out yet?), you still have access to the office. If you need high speed internet that's more reliable than eduroam, it's a good option. Also the free coffee is good.
- You still need to do Pebble Pad entries during exam season, and after the Gateway. Yes, it's bullshit. The bag is worth it, though.